Lecture 3: Regularization and Model Stability

Ivy Yang[1]

[1]Economics & Finance, Business School, Southern Illinois University Edwardsville

1 From Regression to Stable Forecasts

1.1 What this lecture adds

Lecture 1 built simple forecasting rules; Lecture 2 separated model fitting, validation and an honest final test. Here we ask how to fit a useful rule when several inputs overlap or there are many candidate inputs. We compare fitting methods on the same forecasting task, measuring future errors as well as sensitivity to the training observations.

1.2 A regression prediction

A regression predicts an outcome by combining input values. A predictor is one of those inputs. A coefficient is the fitted weight on an input; its interpretation holds the other model inputs fixed. The intercept is the starting level before adding the weighted input contributions. A positive coefficient associates a higher input with a higher prediction; a negative coefficient associates it with a lower prediction. Its magnitude is its size. These weights describe a fitted relationship. A causal effect is the change in an outcome produced by changing an input; fitted weights alone do not establish that effect.

The observed outcome is the recorded value. A fitted value is the model's estimate for an observation used in fitting. A residual is the observed outcome minus its fitted value. Ordinary least squares, shortened to OLS, chooses the intercept and coefficients to minimize the sum of squared training residuals.

1.3 Two meanings of stability

Stability concerns sensitivity to a change in the data used for fitting. Coefficient stability means that modest changes in the training observations do not produce large, unexplained changes in estimated coefficients. Prediction stability concerns changes in the predictions themselves. The two can differ: coefficients on similar inputs can move in opposite directions while their combined prediction changes little.

1.4 Bias, variance and the goal of fitting

Bias here means systematic error in a fitted relationship when averaging over hypothetical repeated training samples. Variance of an estimator measures how much its fitted values

or coefficients vary across those samples. These meanings differ from simply averaging one model's realized forecast errors. A model may accept a more restrictive relationship to reduce its sensitivity to accidental training observations. Whether the resulting trade-off helps needs validation.

1.5 Inputs and interpretation

Interpretability means being able to explain how a model uses its inputs to produce a prediction. A predictive signal is information in an input that helps predict the target. A high-dimensional candidate set contains many inputs relative to the number of observations available for fitting. In the optional finance application, MSFT and AAPL identify Microsoft and Apple stocks. The S&P 500 is a stock-market index; VIX is an index measuring expected market volatility, or the size of market return fluctuations. A return is a price change expressed relative to its starting price; a change is the later value minus the earlier value. A lagged input uses a value already observed before the target period.

2 Why Stability Matters in Forecasting

Definition: Coefficient stability

How consistently estimated coefficients keep their sign and magnitude across nearby training windows.

Definition: Bias–variance trade-off

A change in fitting restrictions can move the average fitted prediction away from the underlying relationship (bias) while reducing variation across training samples (variance). Whether the combined change improves future accuracy must be measured. More restriction can also underfit.

Forecast accuracy and coefficient stability answer different questions. Accuracy measures how close the forecasts are; stability measures whether nearby estimation windows produce comparable model behavior. Model choice should examine both.

Key idea. The same target and predictor set can produce very different coefficients when the training window changes.

Forecasting context.

- Scenario: predict next-day S&P 500 return using lagged returns of MSFT, AAPL, and VIX.
- In January, model coefficients look reasonable; in March, refitting on a new window changes signs and magnitudes.

- Large sign or magnitude changes are a forecasting risk signal, even when the code is correct.
- Distinguish accuracy (RMSE) from reliability (coefficient stability and interpretability over time).

Concept contrast.

Metric	What it measures	Main failure mode
Training squared error	Fit to observations used for estimation	Overfitting / noise memorization
Out-of-sample RMSE	Forecast performance on unseen data	Regime change / little predictive information
Coefficient stability	Consistency of fitted coefficient values	Overlapping inputs / sample sensitivity

2.1 A small comparison we can calculate

Consider two fitted models with the same intercept of 10. Model A gives weights 2 and 0 to two nearly identical inputs. Model B gives weights 0 and 2. For inputs 5 and 5, both predict 20. If a future case has inputs 5 and 8, Model A predicts 20 and Model B predicts 26. Individual weights and predictions answer different stability questions. All numbers in this example are illustrative, not measured financial results.

Implication. If OLS is sensitive to small window shifts, charging for large fitted weights is a candidate remedy; this is called regularization and is developed below. Large changes may also reflect a genuinely changing relationship; shrinking coefficients does not solve that problem by itself.

3 Multicollinearity and Coefficient Volatility

Definition: Multicollinearity

Multicollinearity means that one predictor can be closely approximated by a weighted sum of other predictors. The model then has difficulty separating their individual contributions. Two inputs moving almost together, or in opposite directions, are a simple case.

3.1 Overlapping information

Correlation measures the direction and strength of a straight-line relationship between two quantities. A value near 1 indicates that they move closely together; a value near -1 indicates that one tends to rise as the other falls. Pairwise means examining two variables at a time. Multicollinearity can also involve several inputs jointly, even without very strong pairwise correlation.

Core intuition. When predictors carry overlapping information, OLS struggles to allocate contribution cleanly.

- Analogy: if two siblings always submit group work together, it is difficult to assign separate credit.
- In forecasting data, highly correlated predictors create the same attribution problem.
- Small noise in one sample can produce large coefficient changes in another sample.

3.2 The same fit can have different weights

Imagine that two inputs are identical. We can give them weights 2 and 0, 1 and 1, or 3 and -1: all three pairs produce the same combined contribution. With nearly identical inputs, their predictions may be close even though the fitted weights are very different. A new observation where the inputs diverge can expose that difference.

Why separating the inputs is difficult. If two inputs usually move together, many combinations of their coefficients can produce nearly the same fitted prediction. Changing a few observations may cause OLS to choose a very different combination. Look at the prediction path as well as the individual coefficients before deciding that forecast behavior has become unstable.

Forecasting implication.

- Volatile coefficients can produce sensitive forecasts, especially for input combinations unlike those in training. They do not automatically imply volatile predictions.
- A model can look acceptable in-sample but still be unreliable in production use.
- An added charge for large coefficient sizes can make the fit less sensitive; compare it with OLS on the same future dates.

4 Ridge Regression: Stability-Oriented Shrinkage

4.1 Loss, penalty and strength

Definition: Regularization

Regularization adds a cost for model complexity while fitting. A loss function scores how badly a model fits or predicts; a penalty is the extra cost placed on a complex or large-coefficient solution. Shrinkage means pulling estimated coefficients toward zero.

The penalty strength, written as λ, controls how much weight the fitting procedure gives that extra cost. It is a hyperparameter: a setting chosen outside the coefficient-fitting step. Choose it using validation data rather than the final test sample.

4.2 Read the prediction before the fitting score

Definition: Ridge regression

Fit a regression with an additional cost equal to the total squared slope weights, multiplied by a chosen penalty strength. This discourages large weights.

Core idea. Fit and coefficient size are balanced in a single score. The squared-weight penalty is called L_2; the subscript identifies the squared-size rule.

Write the regression prediction for training observation i as

$$\hat{y}_i = a + \sum_{j=1}^{p} \beta_j x_{ij}.$$

Here i labels an observation, j labels a predictor, and p is the number of predictors. The input x_{ij} is predictor j for observation i. The intercept a is the starting level; β_j controls the contribution from that input. The sum adds those contributions. All predictors used for a forecast must be available at its origin.

4.3 Fit and coefficient cost

Ridge chooses coefficients to make the following score small:

$$\underbrace{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}_{\text{training error}} + \lambda \underbrace{\sum_{j=1}^{p}\beta_j^2}_{\text{coefficient penalty}}.$$

The first part measures fit across n training observations. The second adds squared coefficient sizes; large coefficients cost more. The intercept is not penalized here. A larger λ asks for more shrinkage, but excessive shrinkage can erase useful relationships. Software packages may scale the two parts differently, so the numerical setting must be read with the package's definition.

- $\lambda = 0$: ridge equals OLS.
- Larger λ: stronger shrinkage, often less sensitivity to training noise and potentially more bias.
- Business interpretation: assess whether less sensitivity is worth the additional restriction for the forecasting task.
- Hyperparameter tuning must be done using validation RMSE, not in-sample fit.

4.4 An explicit fit–penalty comparison

Suppose two candidate fits have mean training squared errors 4 and 5. Their two slope weights are $(2, 0)$ and $(1, 1)$. The squared-weight totals are 4 and 2. At penalty strength 1,

the combined scores are 8 and 7: the second fit wins despite worse training fit. At strength 0.1, the scores are 4.4 and 5.2: the first fit wins. These arithmetic examples compare candidate fits; the actual fitting procedure searches over all allowed coefficients.

Key takeaway. Ridge does not automatically choose a small subset of inputs. It generally retains nonzero weights; a coefficient can still be zero in special data configurations. Increasing the penalty reduces the total squared coefficient size for a fixed fitting problem, but every individual correlated coefficient need not move monotonically toward zero.

5 Lasso Regression: Shrinkage and Sparsity

Definition: Lasso regression (L_1)
Regularization using $\sum_{j=1}^{p} \lvert\beta_j\rvert$, which can shrink some coefficients exactly to zero.

5.1 What an exact zero does

The absolute-size rule is called L_1; it adds coefficient magnitudes without squaring them.

Sparsity means that many coefficients are exactly zero. Variable selection, also called feature selection, means choosing which inputs remain in the model. A feature is another name for an input or predictor. Lasso can perform selection because its absolute-value penalty can produce exact zeros.

5.2 Absolute-value coefficient cost

Lasso chooses coefficients to make this score small:

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

The training-error part is the same as above. In the penalty, absolute values measure coefficient sizes without squaring them. A coefficient set to zero removes that input's contribution from the fitted prediction. This property does not establish that the input has no economic effect.

- Practical value: automatic variable screening when there are many candidate features.
- Communication value: sparse models are easier to explain to non-technical stakeholders.
- Caution: in finance, many signals are weak; lasso may discard useful but small effects.
- Stability caution: selected variables may flip across windows.

6 Ridge vs. Lasso: Forecasting Decision Rules

6.1 Compare candidates rather than declaring a winner

A coefficient path displays fitted weights against penalty strength for a fixed data set. A stability plot instead compares estimates across different fitting samples. Moving along a penalty path and moving the training window are different experiments.

Decision principle. Compare forecasting error, sensitivity and simplicity on the same task. Neither penalty guarantees a better forecast. A sparse signal means useful predictive information is concentrated in relatively few candidate inputs.

	Ridge	Lasso
Penalty	$L_2 : \sum \beta_j^2$	$L_1 : \sum \lvert\beta_j\rvert$
Shrinkage style	Smooth and continuous	Sharp, can hit zero
Variable selection	No	Yes
Best use case	Many correlated predictors, all likely relevant	High-dimensional setting with sparse true signal
Stability across windows	Often improved relative to OLS	Selection may flip between similar inputs

- When many correlated inputs may carry shared signal, ridge is a useful candidate.
- When a compact set of inputs matters, evaluate lasso and check whether its selected set persists.
- Retain a simple baseline if neither regularized model improves the measured result. Financial returns may contain little usable signal.

6.2 Optional extension: elastic net

A third option, elastic net, combines both penalties:

$$\text{score} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda_1 \sum_{j=1}^{p} |\beta_j| + \lambda_2 \sum_{j=1}^{p} \beta_j^2.$$

Elastic net can combine exact-zero selection with shrinkage of groups of related inputs. These properties do not guarantee better validation forecasts. Here λ_1 weights the absolute-value penalty and λ_2 weights the squared-coefficient penalty. In practice: start with ridge or lasso; consider elastic net only when neither alone performs satisfactorily on validation data.

7 Implementation Workflow: OLS, Ridge, and Lasso

7.1 Keep time order and preserve the final test

Experimental protocol.

1. Training set: first 70% of observations for model fitting.

2. Validation set: next 15% for selecting λ (ridge and lasso).

3. Test set: final 15% used exactly once for final comparison.

The 70/15/15 split is an illustrative allocation, not a universal rule. Choose dates that leave enough observations and match the forecasting horizon. At each date, inputs must already be available.

7.2 Why input units change the penalty

Preprocessing means transforming raw observations into the inputs used for fitting. Standardization subtracts a predictor's training mean and divides by its training standard deviation. For an input, variance is the average squared distance from its mean. A standard deviation is the square root of that variance and measures spread in the original units. A scaler stores the training means and standard deviations and applies this transformation. The result measures how many training standard deviations an input lies above or below its training mean. A predictor with zero training variance needs separate handling rather than division by zero.

Imagine measuring the same distance in meters or centimeters. Multiplying an input by 100 lets its coefficient divide by 100 with no change in prediction. The squared-coefficient penalty then divides by 10,000; the absolute-value penalty divides by 100. Without a consistent scale, the fitting rule penalizes units as well as model behavior.

With training mean m_j and training standard deviation $s_j > 0$, the transformed input is

$$z_{ij} = \frac{x_{ij} - m_j}{s_j}.$$

The numerator measures distance from the training average. Dividing by the training spread expresses that distance in comparable units. A new observation uses the same stored training mean and spread.

Critical preprocessing step: standardize predictors first. Ridge and lasso penalties are applied to raw coefficient values. If predictors are on very different scales (e.g., VIX on a 10–80 range, a binary indicator, meaning an input that takes only values 0 and 1), the penalty treats them unequally. Use a consistent predictor scale before comparing ridge or lasso penalties. Fit the standardization on the current training window only, then apply that same transformation to the validation or test inputs. Refit it within each training fold when performing repeated validation. In sklearn, use StandardScaler() before Ridge or Lasso. With an intercept and a uniquely identified least-squares fit, a change of predictor units does not change OLS fitted predictions. Standardizing all candidates is convenient for a common workflow and comparable coefficient displays; it does not make unlike data sets comparable.

7.3 Choose strength using validation only

How to select λ in practice.

1. A grid is a list of candidate settings to try. Define a range of penalty strengths from weak to strong. The numerical range depends on the software objective and data scale.

2. For each candidate λ, fit the model on the training set and compute RMSE on the validation set only.

3. Select the λ that minimizes validation RMSE.

4. Keep the chosen setting fixed. Re-fit the entire preprocessing-and-model procedure on train+validation before the later test period. Re-estimate scaling from this enlarged fitting sample; do not include any test observations.

5. Evaluate once on the test set for final reporting.

Never use the test set to pick λ: that leaks evaluation information into model selection. A validation curve need not be U-shaped. If its best score lies at an edge, inspect a wider sensible range using validation alone. After inspecting test results, do not keep changing the model and call the same test an untouched final evaluation.

7.4 Report what was actually measured

A selection frequency is the fraction of repeated fits in which an input has a nonzero coefficient. This describes selection behavior across fitting samples; it does not measure a causal effect.

Modeling sequence.

- Standardize predictors; build OLS baseline, one ridge model, and one lasso model on identical features.

- Tune λ only on validation data using the grid procedure above.

- Compare test RMSE and rolling-window coefficient stability.

- Write a short recommendation memo in manager-facing language.

Model recommendation memo template. A reproducible comparison can be run again from the same input data and documented steps. Fill in the measured results of your run; do not treat the model name as evidence that it won.

- Task: Forecast next-day S&P 500 return using stock returns already observed at the origin and the observed VIX change.

- Evidence: Report measured errors on identical test dates, together with prediction and

coefficient sensitivity. Do not assume in advance which model wins.

- Recommendation: Name the preferred model only after comparing measured error, sensitivity and the operating need.
- Exception: Explain any loss of accuracy accepted for a smaller input set; report selection changes across windows.

8 Knowledge Checklist

1. Why can a model with good in-sample fit still fail in forecasting practice?
   In-sample fit measures historical description quality, not future robustness. Overfitting and coefficient instability can both create weak out-of-sample performance.

2. How does regularization improve model stability under multicollinearity?
   Penalties constrain coefficient magnitude and can reduce sensitivity to overlapping predictor information. Check validation forecasts and stability rather than assuming the penalty always helps.

3. What is the core trade-off between ridge and lasso?
   Ridge adds squared-coefficient cost and often reduces sensitivity when predictors share information. Lasso adds sparsity and variable selection but can be less stable across rolling windows.

4. How do forecasting goals change the preferred regularization method?
   Use ridge as a candidate for shared-input shrinkage and lasso as a candidate for compact selection. The final recommendation still needs validation and sensitivity evidence.

9 Interview Questions This Lecture Prepares You To Answer

- "Why did you regularize your model?"
  "The predictors are correlated, and OLS coefficients were unstable across nearby windows. I compared regularized candidates with OLS using validation forecasts and repeated-window sensitivity, then reported whether they actually helped."

- "What do you gain and lose by using lasso or ridge?"
  "Ridge adds a squared-weight penalty and generally retains all inputs. Lasso gives sparse, easier-to-explain models but may drop weak yet useful signals and can flip selected features over time."

- "Why is stability important in forecasting?"
  "A model that changes its story every re-estimation cycle is difficult to trust in risk committees and business planning, even if average RMSE is acceptable."

10 Assignment Connection

Coding Assignment 2 directly extends this lecture. The theory examples illustrate mechanisms; they are not empirical evidence that a named model beats another on financial data. Students are expected to submit:

- one reproducible OLS/ridge/lasso comparison,
- one validation-driven hyperparameter decision,
- one short manager-style model recommendation,
- and one discussion of stability vs. interpretability trade-offs.